Exam

Electricity and Electronics EBN 111

13 June 2022

Test Information			
Maximum marks:	93	Full marks:	90
Duration of test:	200 min	Open/Closed book:	Closed
Total number of pages (this page included)		32	

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ECSA GA assessment

The following ECSA graduate attributes are assessed at entry level in this question paper: GA1 and GA2.

Important

1. The examination regulations of the University of Pretoria apply.
2. The departmental rules relevant to electronically graded assessments apply.
3. Write your answers on the OCR forms provided. The question number **01** in this question paper corresponds to Row 01 of the OCR form, etc.
4. Read the instructions on the OCR form carefully.
5. Answers must include units where applicable.
6. Final answers must be written as real numbers and not as fractions. Use at least 4 significant figures to write numbers, including complex numbers.

INSTRUCTIONS

Do STEP 0 in the first 5 minutes of the session. (-5 to 0 min)

Do STEP 1 in the next 90 minutes of the test. (0-180 min)

Do STEP 2 within the next 20 minutes. (180-200 min)

STEP 0 (-5 min to 0 min): Complete the front page of the OCR form, and also fill in your name and student number on additional pages of the OCR forms.

STEP 1 (0 min to 180 min): Do your own calculations on the question paper. Write down the answers on the PRACTICE ANSWER SHEET. Be sure to include the units where applicable.

STEP 2 (180 min to 200 min): Carefully copy the answers from the PRACTICE ANSWER SHEET to the OCR form.

- The Assessment ID is: **2022EBN111E01**
- Use a mechanical pencil with 0.5 mm HB lead to complete the form.
- Each cell should only have one character, and characters may not touch or cross the cell.
- **Do not use commas!** Make use of full stops as decimal points.

Section 1

Question 1 [2]

An electric motor can deliver 50 kJ in 2 minutes. Find the annual cost (in Rand) of running the motor if it is used for 5 hours a day, 300 days a year. The cost of electricity is R1/kW h.

01	Annual cost (in Rand) [2]

$$P = \frac{\Delta W}{\Delta T} = \frac{50\,kJ}{2\,min \times 60\,s/min} = 0.4167\ kW$$

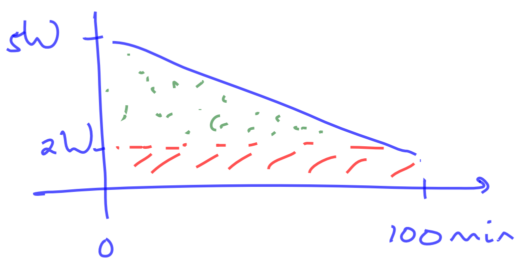
$$Cost = 0.4167\,kW \times 5h \times 300 \times R1/kWh$$

$$= R\,625$$

Question 2 [2]

A 9 V battery is used to power a remote controlled car. The power delivery from the battery falls linearly from 5 W to 2 W in 100 minutes. Determine the total energy W_T used by the car.

02	W_T [2]



$$W_T = \int p\,dt = 2 \times 100\,min \times 60\,s/min$$

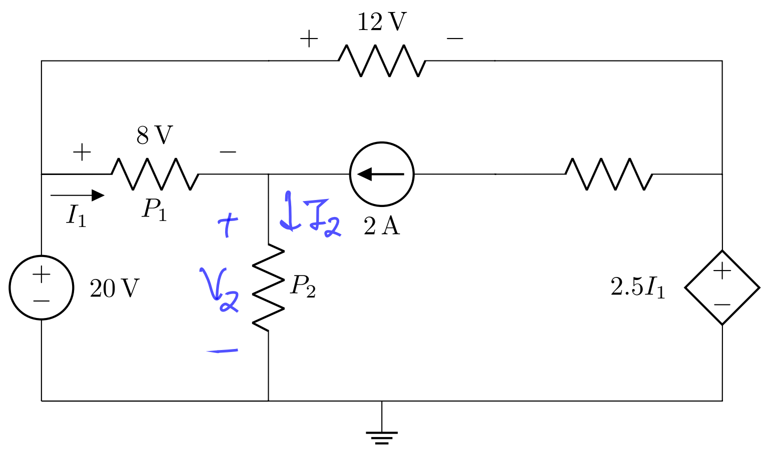
$$+ \frac{1}{2}(5-2) \times 100\,min \times 60\,s/min$$

$$= 21\,kJ$$

Questions 3 to 4 [4]

For the circuit shown below, determine P_1 and P_2.

(There is sufficient information in the circuit to solve the problem.)



03	P_1 [2]
04	P_2 [2]

03 KVL: $-20 + 12 + 2.5 I_1 = 0$

$I_1 = 3.2\,A$

$P_1 = 3.2 \times 8 = 25.6\,W$

04 KCL: $I_2 = 3.2 + 2 = 5.2\,A$

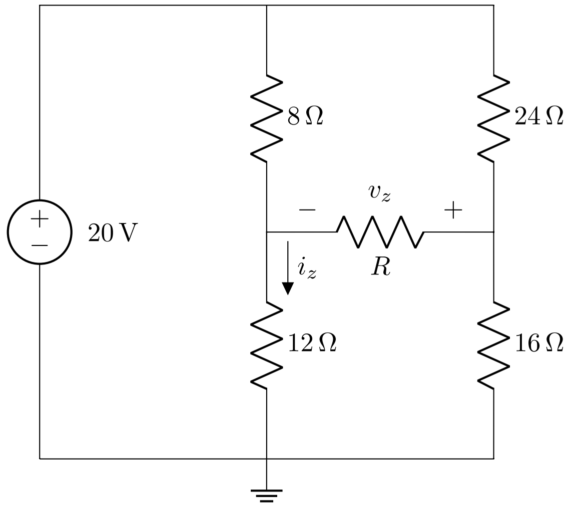
KVL: $-20 + 8 + V_2 = 0$

$V_2 = 12\,V$

$P_2 = 5.2 \times 12$

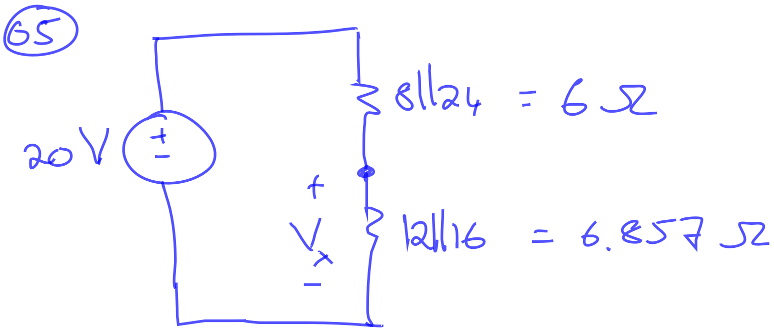
$= 62.4\,W$

Questions 5 to 6 [4]	
For the circuit below:	



05	If $R = 0\,\Omega$, determine i_z. [2]
06	If $R \to \infty$, determine v_z. [2]

05



$V_x = \frac{6.857 \times 20}{6 + 6.857}$

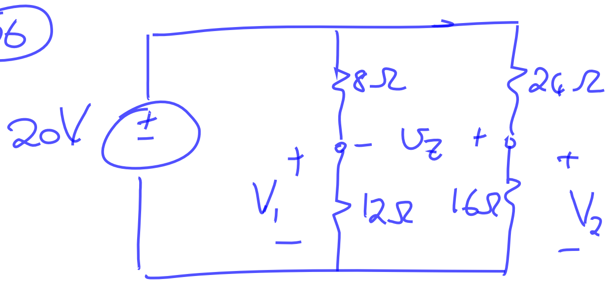
$= 10.667\ V$

$i_z = \frac{10.667}{6.\cancel{857}}$ 12

$= 1.\cancel{5}56\ A$

0.8889A

06



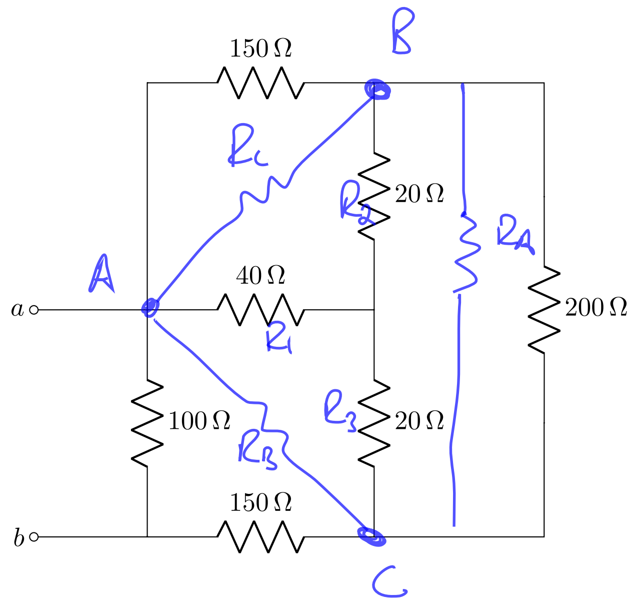
$V_1 = \frac{20 \times 12}{12 + 8} = 12V$

$V_2 = \frac{20 \times 16}{16 + 24} = 8V$

$v_z = V_2 - V_1$

$= -4V$

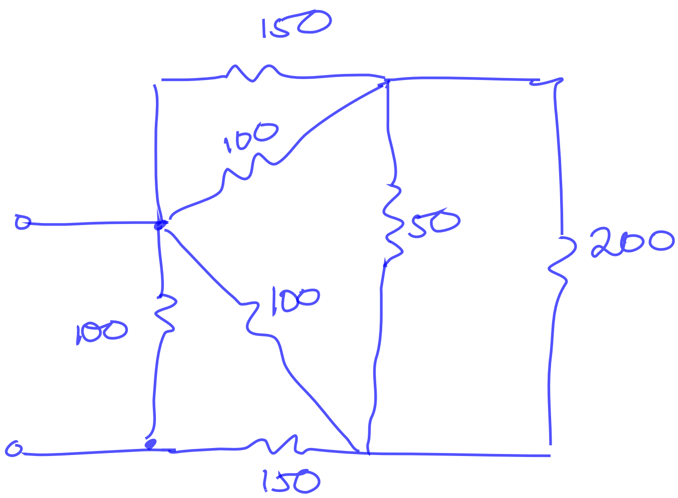
Question 7 [2]
Determine the equivalent resistance R_{ab}.



07	R_{ab} [2]

$$R_A = \frac{20\times20 + 40\times20 + 40\times20}{40} = \frac{2000}{40} = 50\,\Omega$$

$$R_B = R_C = \frac{2000}{20} = 100\,\Omega$$



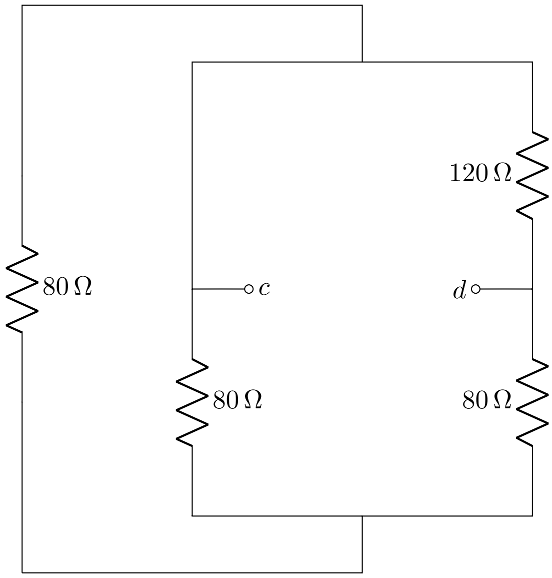
$$R_{ab} = 100 \,||\, (150 + 100 \,||\, (100||150 + 200||50))$$

$$= 100 \,||\, (150 + 100 \,||\, (60 + 40))$$

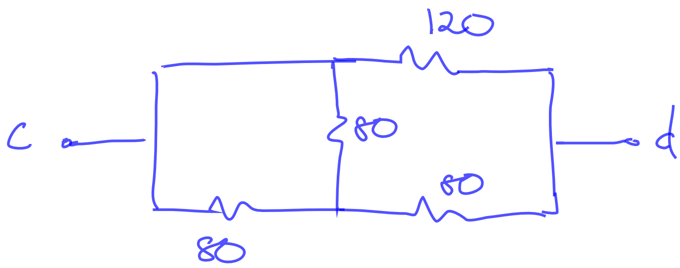
$$= 100 \,||\, (150 + 50) = 100 \,||\, 200 = \underline{66.667\,\Omega}$$

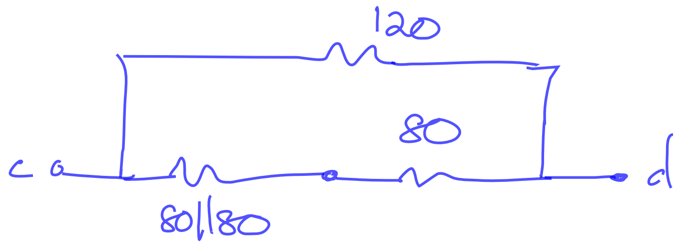
Question 8 [2]

Determine the equivalent resistance R_{cd}.



08	R_{cd} [2]





$$R_{cd} = 120 \parallel (40 + 80)$$

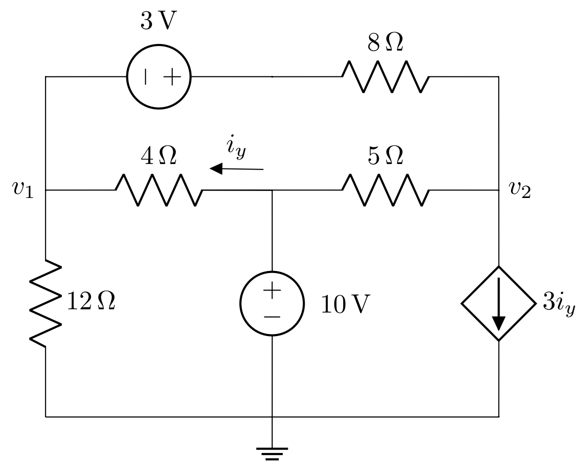
$$= 60\,\Omega$$

Questions 9 to 12 [4]

Use nodal analysis to set up two equations in the form:

$$av_1 + bv_2 = 51,$$
$$cv_1 + dv_2 = -205.$$

Calculate the unknown coefficients a, b, c, and d.



09	a [1]
10	b [1]
11	c [1]
12	d [1]

$$\frac{v_1}{12} + \frac{v_1 - 10}{4} + \frac{v_1 + 3 - v_2}{8} = 0$$

$(\times 24)$ $\quad 2v_1 + 6v_1 - 60 + 3v_1 + 9 - 3v_2 = 0$

$$11v_1 - 3v_2 = 51$$

$$3i_y + \frac{v_2 - 10}{5} + \frac{v_2 - 3 - v_1}{8} = 0$$

$(\times 40)$ $\quad 120i_y + 8v_2 - 80 + 5v_2 - 15 - 5v_1 = 0$

$$120i_y - 5v_1 + 13v_2 = 95$$

but $i_y = \frac{10 - v_1}{4}$

$$\therefore 120\left(\frac{10 - v_1}{4}\right) - 5v_1 + 13v_2 = 95$$

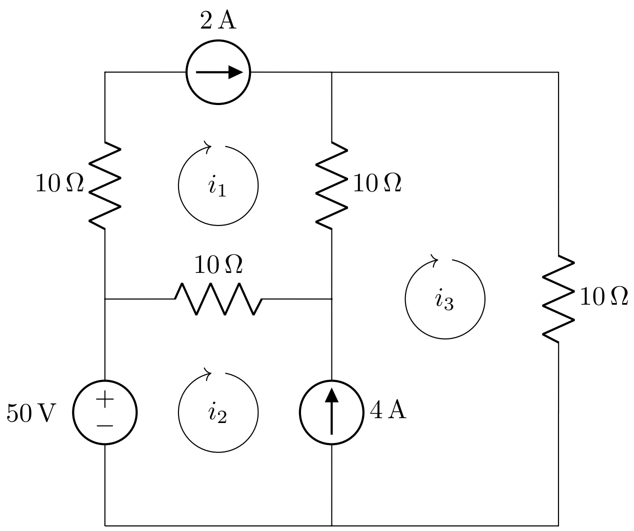
$$\hookrightarrow \quad -35v_1 + 13v_2 = -205$$

Questions 13 to 16 [4]

Use mesh analysis to set up two equations in the form:

$$ei_2 + fi_3 = 90,$$
$$-i_2 + gi_3 = h.$$

Calculate the unknown coefficients e, f, g, and h.



13	e [1]
14	f [1]
15	g [1]
16	h [1]

* $i_1 = 2A$

Supermesh *

$-50 + 10i_2 - 10i_1 + 10i_3 - 10i_1 + 10i_3 = 0$

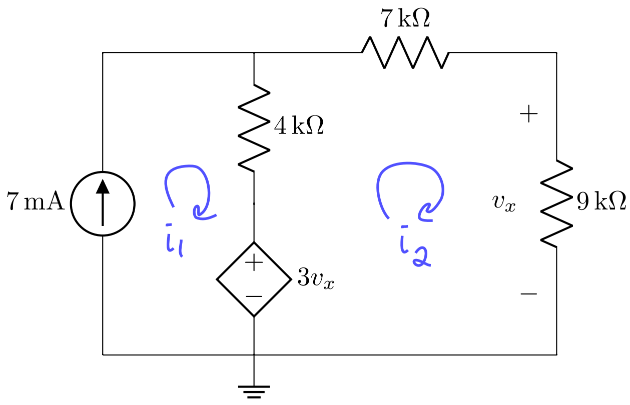
$10i_2 + 20i_3 = 50 + 20(2) = 90$

KCL: *

$-i_2 + i_3 = 4$

Question 17 [2]

Determine v_x using either nodal or mesh analysis.



17	v_x [2]

* Mesh:

* $i_1 = 7mA$

* $-3v_x + i_2(4k + 7k + 9k) - i_1(4k) = 0$

$(20k)\, i_2 - 3v_x = 4k \times 7m$

but $i_2 = \frac{v_x}{9k}$

$\therefore \frac{20k}{9k} v_x - 3v_x = 28$

$v_x = \frac{28}{\frac{20}{9} - 3} = -36\,V$

* Nodal

$7m = \frac{V_1 - 3v_x}{4k} + \frac{V_1}{16k}$

$(\times 16k)$ $112 = 4V_1 - 12v_x + V_1 = 5V_1 - 12v_x$

but $v_x = \frac{9V_1}{16} \rightsquigarrow V_1 = \frac{16v_x}{9}$

$\hookrightarrow$ $112 = \left(5\left(\frac{16}{9}\right) - 12\right) v_x$

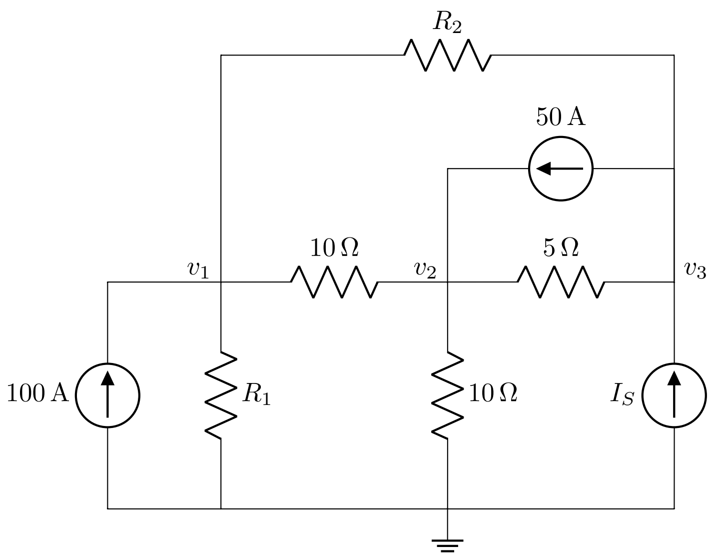
$v_x = -36\,V$

Questions 18 to 20 [3]

The matrix below was constructed by means of inspection in terms of the node voltages for the circuit below:

$$\begin{bmatrix} 0.35 & -0.1 & -0.05 \\ -0.1 & 0.4 & -0.2 \\ -0.05 & -0.2 & 0.25 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 100 \\ 50 \\ -10 \end{bmatrix}$$

Determine R_1, R_2, and I_S.



18	R_1 [1]
19	R_2 [1]
20	I_S [1]

$* \; R_2 = \frac{1}{0.05} = \underline{20\Omega}$

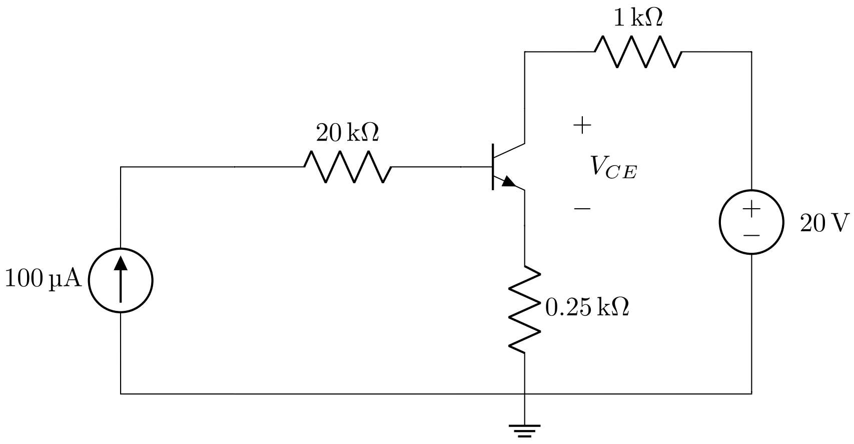
$* \; R_1 = (0.35 - 0.05 - 0.1)^{-1} = \frac{1}{0.2} = \underline{5\Omega}$

$* \; I_s - 50 = -10$

$I_s = \underline{40\text{ A}}$

Question 21 [2]

Assume $I_B = 100\,\mu\text{A}$ and $\beta = 99$, determine V_{CE}.



21	V_{CE} [2]

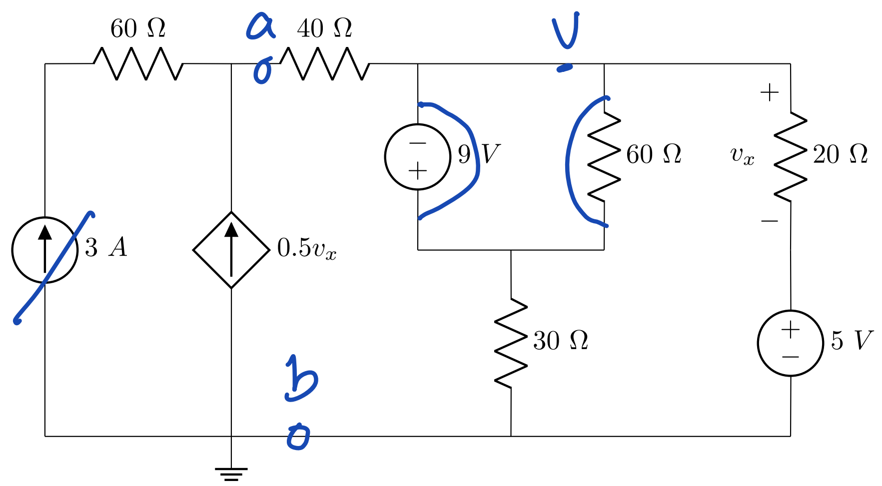
$$V_{CE} = 20 - 1k(99 \times 100\mu) - 0.25k(99+1)(100\mu)$$

$$= 20 - 9.9 - 2.5$$

$$= 7.6\,V$$

Total Section 1: [31]

Section 2

Questions 22 to 23 [4]	
Use superposition to determine: 1) v_x^1, the exclusive contribution from the 5 V independent voltage source to v_x; and 2) v_x^2, the exclusive contribution from the 3 A independent current source to v_x.	
	
22	v_x^1 [2]
23	v_x^2 [2]

22. When only 5V is functioning, use nodal analysis

$$-0.5V_x + \frac{V}{30} + \frac{V-5}{20} = 0, \quad V_x = V-5$$

$$-0.5(V-5) + \frac{V}{30} + \frac{V-5}{20} = 0, \quad \times 60$$

$$-30(V-5) + 2V + 3V - 15 = 0$$

$$\Rightarrow \quad -25V = -135 \Rightarrow V = 5.4\,V$$

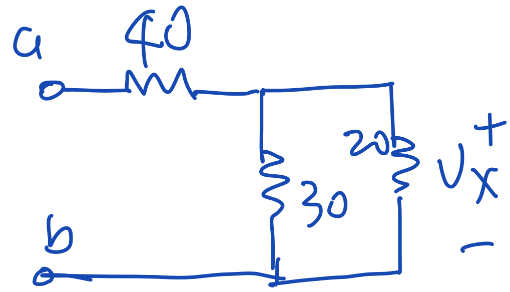
$$V_x^1 = V - 5 = 0.4\,V$$

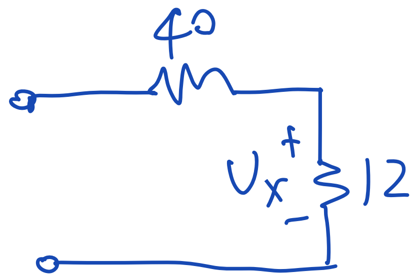
23. When only the 3A is functioning, use nodal analysis:

$$V_x = 12(3 + 0.5V_x)$$

$$\Rightarrow 5V_x = -36$$

$$\Rightarrow V_x^{2} = -7.2\,V.$$

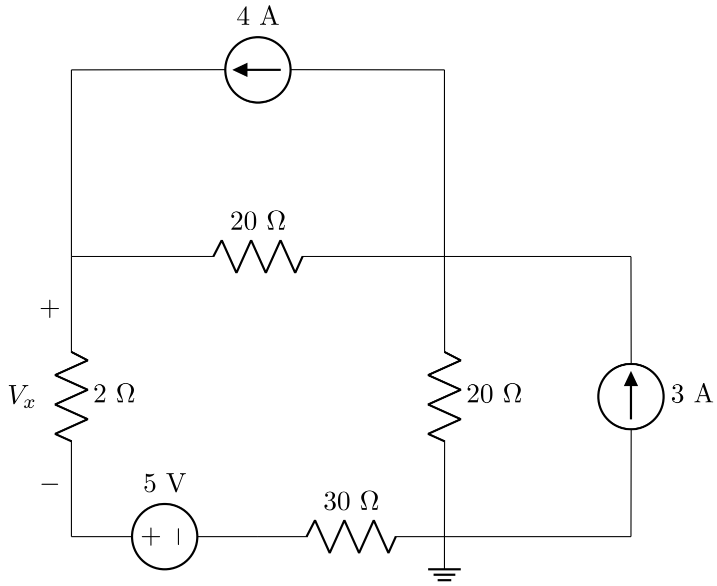




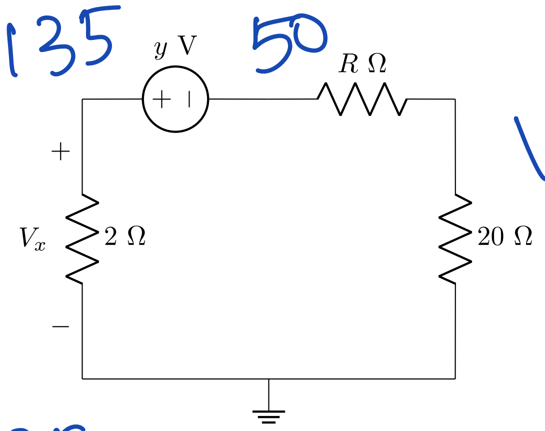
Questions 24 to 25 [3]

Use source transformation to transform Circuit A to Circuit B.

Circuit A:



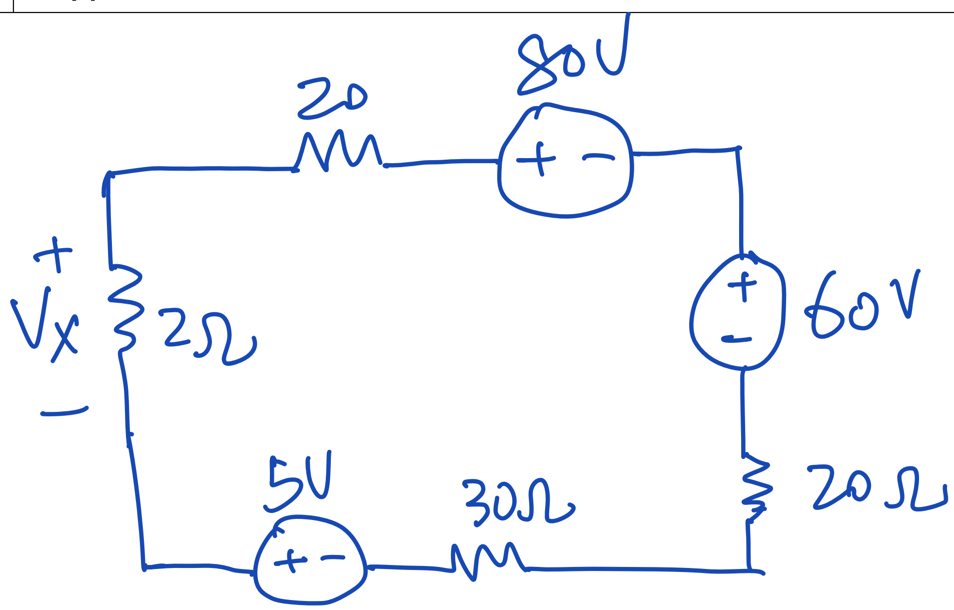
Circuit B:



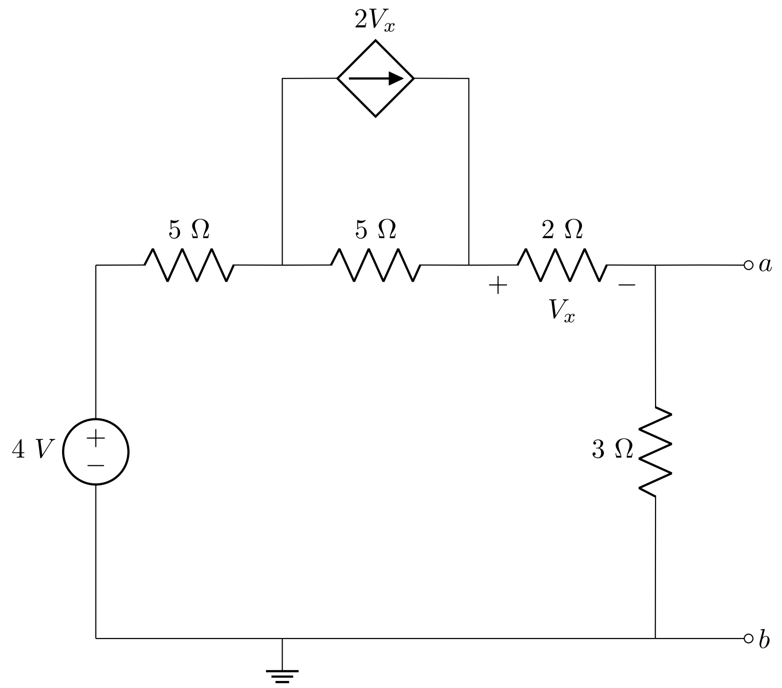
$$V_x = \frac{2 \times 135}{2+70}$$

$$= 3.75\,V.$$

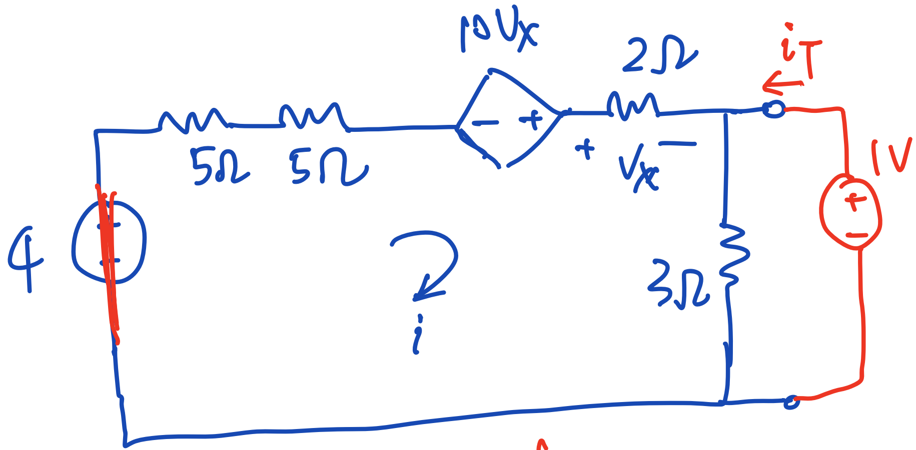
24	y [2]	$y = 135$
25	V_x [1]	



Questions 26 to 27 [4]	
Determine the Thévenin equivalent voltage V_{Th}, and the Thévenin equivalent resistance R_{Th} between terminals $a - b$.	



26	V_{Th} [2]
27	R_{Th} [2]



$V_{Th} = 3i$., $V_x = 2i$

$-4 + 15i - 10V_x = 0$

$-4 + 25i - 20i = 0$

$\Rightarrow 5i = 4 \Rightarrow i = 0.8$

$V_{Th} = 2.4V$.

$i_T = \frac{1}{3} + \frac{1-10V_x}{12}$

$V_x = -2 \cdot \frac{1-10V_x}{12}$

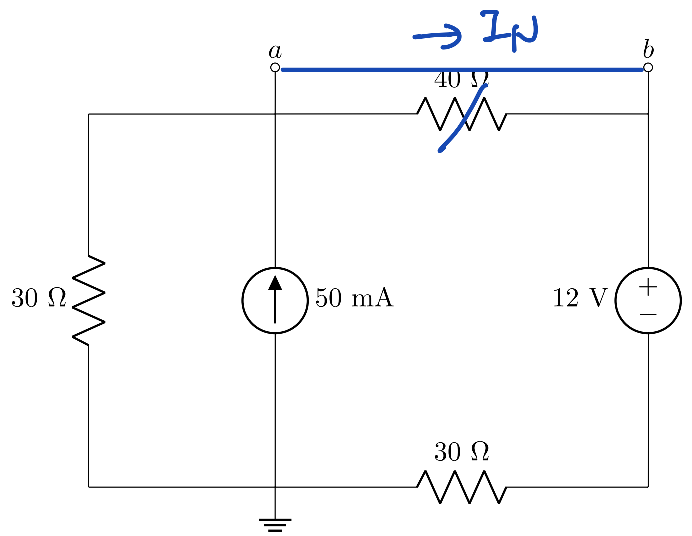
$\Rightarrow 12V_x = -2 + 20V_x$

$\Rightarrow V_x = \frac{1}{4}$, $\Rightarrow i_T = \frac{1}{3} + \frac{1-2.5}{12}$

$= \frac{1}{3} - \frac{1}{8}$

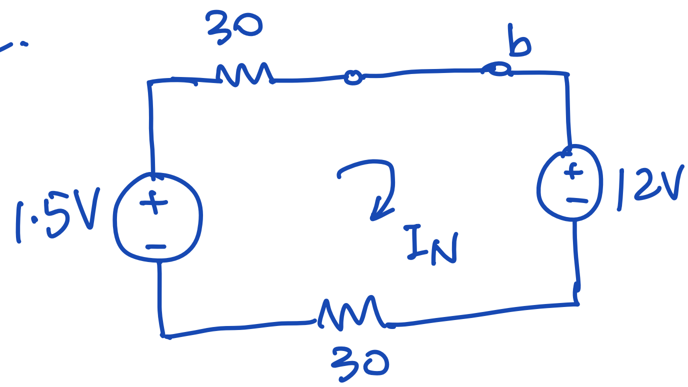
$R_{Th} = \frac{1}{i_T} = 4.8\Omega$

Questions 28 to 29 [4]	
Determine the Norton equivalent current I_N and resistance R_N between terminals $a - b$.	



28	I_N [2]
29	R_N [2] : use source transformation.

$$R_N = 40 \,||\, 60 = 24\ \Omega.$$

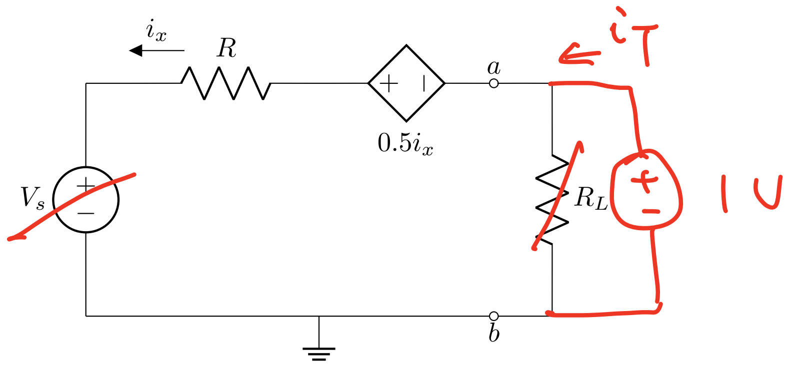


$$I_N = \frac{1.5-12}{60}$$

$$= -0.175.$$

Question 30 [2]

In the circuit below, $i_x = 2$ mA, and the maximum power $P_{max} = 500$ mW is delivered to the load R_L. Determine the Thévenin equivalent voltage V_{TH} of the circuit between terminals $a - b$.



30	V_{TH} [2]

$$P_{max} = \frac{V_{Th}^2}{4R_L} = 0.5$$

when open circuit: $V_{Th} = V_s - 0.5 i_x$, $i_x = 0$

$$\Rightarrow V_{Th} = V_s,$$

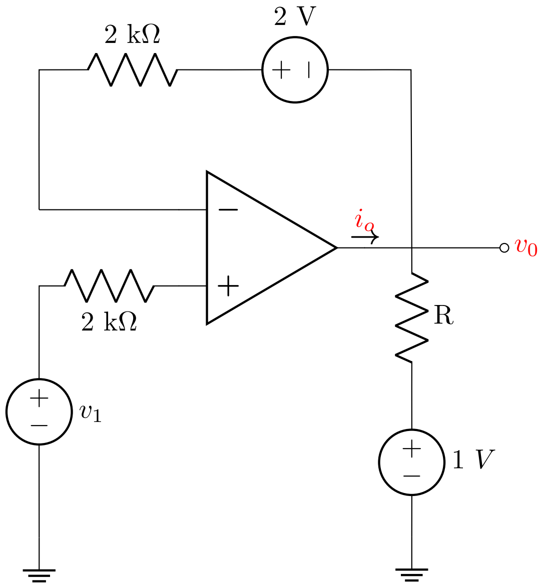
$$i_T = i_x = \frac{1 + 0.5 i_x}{R} \Rightarrow R = \frac{1 + 0.5 i_x}{i_x} = \frac{1}{i_x} + 0.5$$

$$i_x = 1A \Rightarrow R_{Th} = \frac{1}{i_T} = \frac{R}{1 + 0.5 i_x} = 0.5\,\Omega.$$

$= 1\,\Omega$

$$R_{Th} = R_L.$$

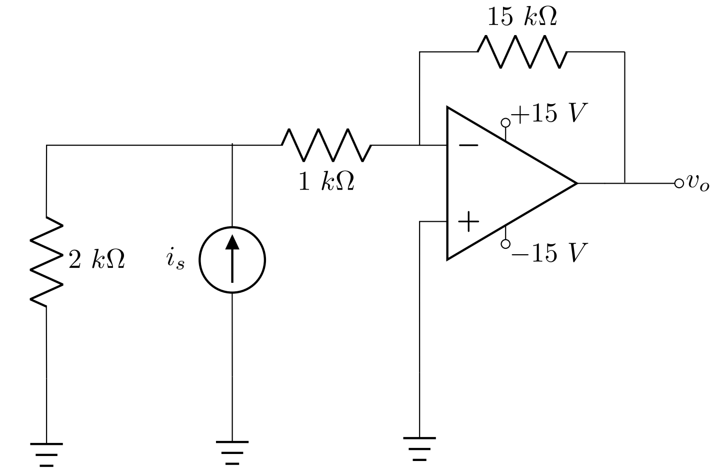
$$\Rightarrow V_{Th} = \sqrt{4 \times 0.5 \times 0.5} = 1V.$$

Questions 31 to 32 [4]	
Assume that the operational amplifier is ideal, determine v_1 and i_o if v_0=5 V and $R = 4k\ \Omega$.	
	
31	v_1 [2]
32	i_o [2]

$$V_0 = V_1 - 2 = 5V \Rightarrow V_1 = 7V.$$

$$i_0 = \frac{V_0 - 1}{R} = \frac{4}{4k} = 1mA.$$

Questions 33 to 34 [4]
The operational amplifier is ideal.



33	Determine v_o when $i_s = 6$ mA. [2]
34	What is the ~~maximum~~ minimum value of i_s ~~that drives~~ such that the operational amplifier ~~to~~ is just about to enter negative saturation? [2]

33. $i = i_s \cdot \frac{2}{2+1} = 6 \cdot \frac{2}{3} = 4\,mA.$

$$= \frac{0 - V_0}{15K} \Rightarrow V_0 = -60\,V.$$

Due to saturation, $V_0 = -15\,V.$

34. $i = i_s \cdot \frac{2}{3} = \frac{0 - V_0}{15K}$

$$\Rightarrow V_0 = -\, i_s \cdot 15K \cdot \frac{2}{3} \le -15$$

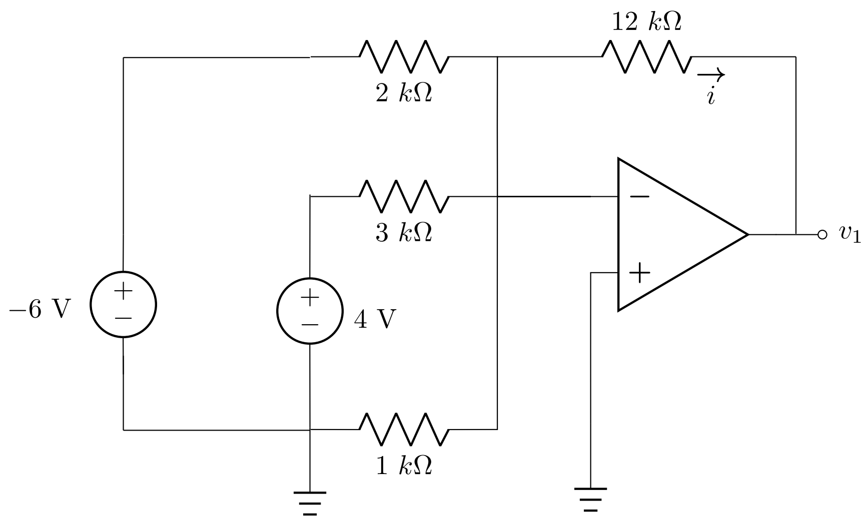
$$\Rightarrow i_s \ge 15 \cdot \frac{3}{2} \cdot \frac{1}{15K}$$

$$i_s \ge 1.5\,mA.$$

$$i_s^{min} = 1.5\,mA$$

Question 35 [2]

Assume that the operational amplifier is ideal, determine v_1.



35	v_1 [2]

35.

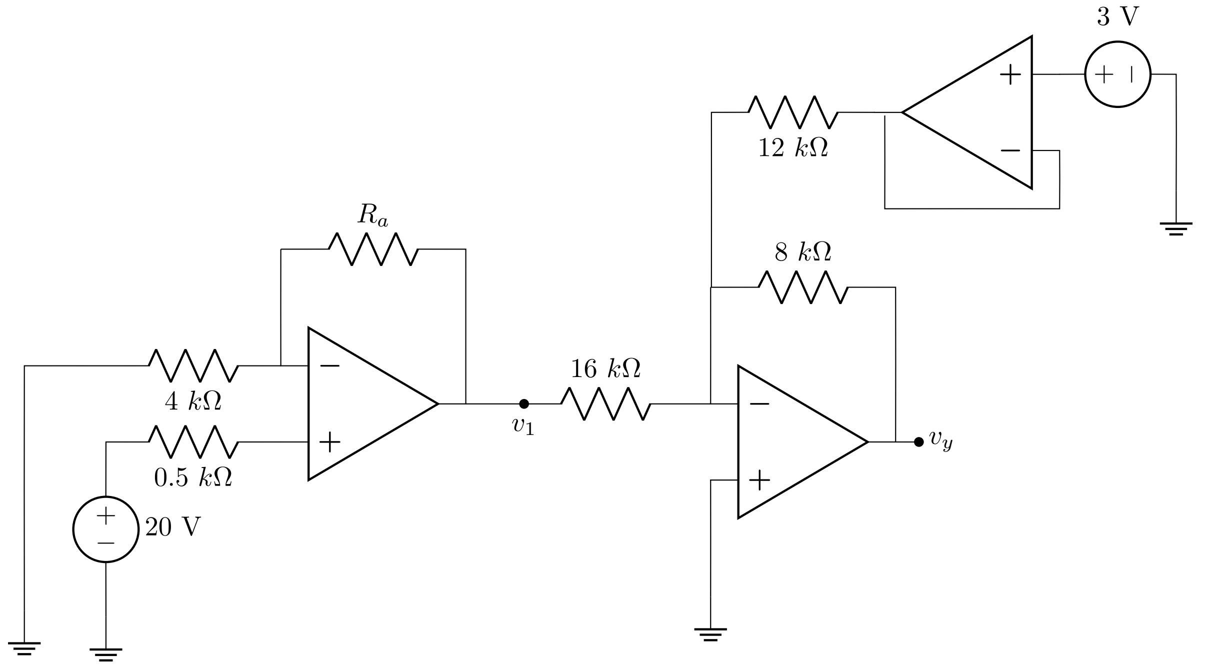
$$\frac{-6}{2K} + \frac{4}{3K} = \frac{0 - V_1}{12K}$$

$$\Rightarrow -36 + 16 = -V_1$$

$$\Rightarrow V_1 = 20V.$$

Questions 36 to 37 [4]

Assume that the operational amplifiers are ideal, calculate R_a and v_y when $v_1 = 30$ V.



36	R_a [2]
37	v_y [2]

36. $30 = 20\left(1 + \frac{R_a}{4k}\right) \Rightarrow 1 + \frac{R_a}{4k} = 1.5$

$\Rightarrow R_a = 2k\,\Omega.$

37. $V_y = -30 \cdot \frac{8}{16} - 3 \cdot \frac{8}{12}$

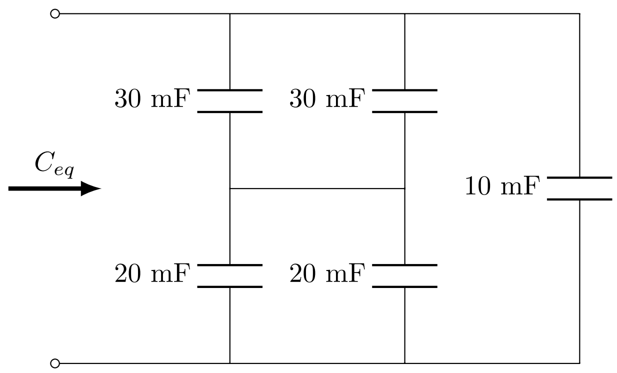
$= -15 - 2 = -17V.$

Total Section 2: [31]

Section 3

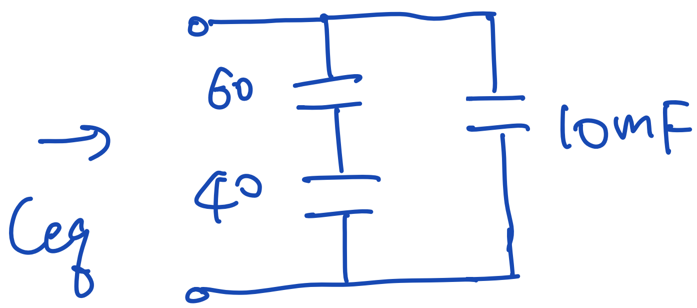
Question 38 [1]

Find the equivalent capacitance C_{eq} in the circuit below.



38	C_{eq} [1]

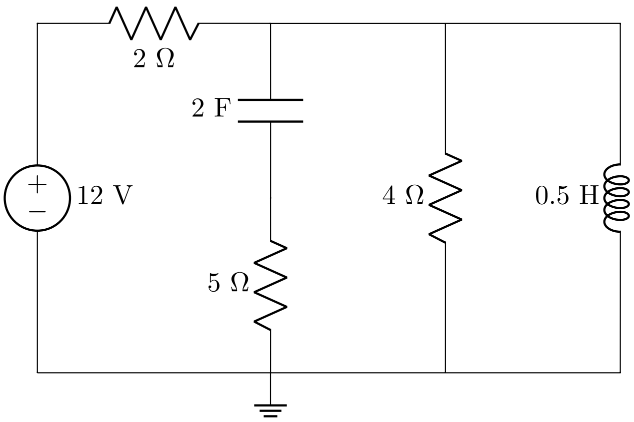
38



$$\frac{60 \times 40}{60+40} = 24\,mF \Rightarrow C_{eq} = 34\,mF.$$

Question 39 [2]

Under dc conditions, the energy stored in the inductor is denoted as w_L and the energy stored in the capacitor is w_C. Calculate the total energy $w_T = w_C + w_L$.



39	w_T [2]

39

$V_C = 0 \Rightarrow w_C = 0$

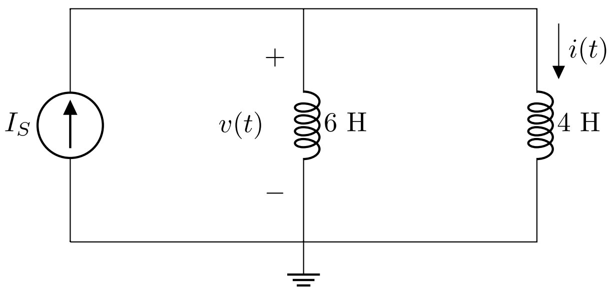
$i_L = \frac{12}{2} = 6\,A$

$\Rightarrow W_L = \frac{1}{2} L i_L^2 = \frac{1}{2} \times 0.5 \times 6^2$

$= 9\,J.$

$W_T = W_L + W_C = 9\,J$

Questions 40 to 41 [4]



40	When $I_S = 4e^{-2t}$ A, determine $v(t)$ at $t = 2$ ms. [2]
41	When $I_S = 2t + 1$ A, determine $i(t)$ at $t = 4$ s, given $i(0) = 2$ A. [2]

$$40: v(t) = L\frac{di}{dt} = (6||4) \times (4e^{-2t})'$$

$$= 2.4 \times 4 \times (-2) e^{-2t}$$

$$= -19.2e^{-2t}.$$

$$v(2ms) = -19.2e^{-2 \times 2 \times 10^{-3}} = -19.12V.$$

$$41: \quad v(t) = L\frac{di}{dt} = 2.4 \cdot 2 = 4.8 \text{ V}.$$

$$i(t) = \frac{1}{L}\int_0^4 4.8\, dt + 2$$

$$= \frac{1}{4} \cdot [4.8t]_0^4 + 2$$

$$= \frac{1}{4} \times 4.8 \times 4 + 2 = 6.8 \text{ A}.$$

Questions 42 to 43 [4]

Consider two sinusoid waveforms $v_1(t)$ and $v_2(t)$, where

$$v_1(t) = -60\cos(30t + 10°), \text{ and}$$
$$v_2(t) = 50\cos(30t + 50°) - 120\sin(30t).$$

The waveforms $v_1(t)$ and $v_2(t)$ can be represented by phasors $\mathbf{V_1}$ and $\mathbf{V_2}$, respectively.

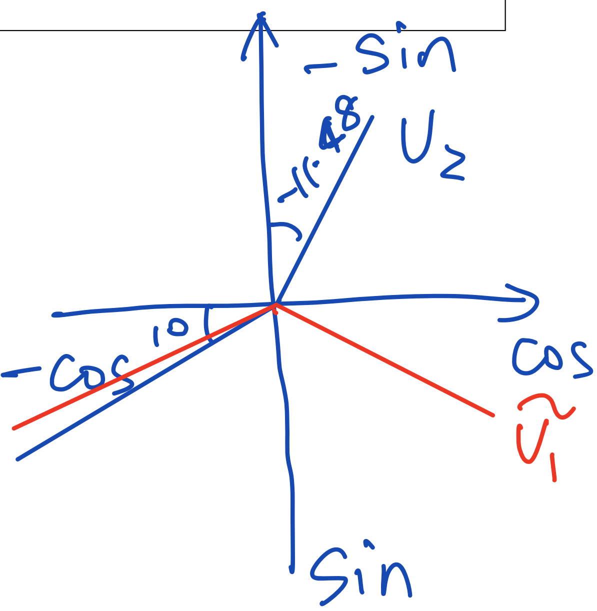
42 Choose the correct statement.

1. v_1 leads v_2 by 60°.
2. v_2 leads v_1 by 60°.
3. v_1 leads v_2 by 101.47°.
4. v_2 leads v_1 by 101.47°.
5. **None** of the above statements are correct.

Write the correct statement number as your answer, choose one from 1, 2, 3, 4, or 5. [2]

$V_1 = 60\angle -170°$

43 Find $\mathbf{V_T} = \mathbf{V_1} + \mathbf{V_2}$. [2]



42: $V_2^1 = 50\angle 50°$

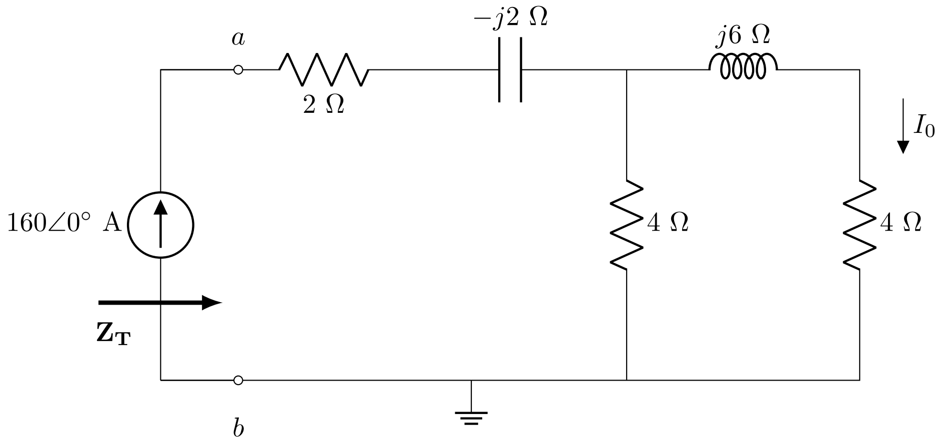
$V_2^2 = 120\angle 90°$

$V_2 = 161.53\angle 78.52°$

Rotate V_1, V_2 counter clockwisely by 120°, then

V_1 leads V_2 by 111.48°.

43: $V_T = 150.32\angle 100.33°$.

Questions 44 to 45 [4]	
	
44	Find $\mathbf{Z_T}$ with respect to terminals $a-b$. [2]
45	Find $\mathbf{I_0}$. [2]

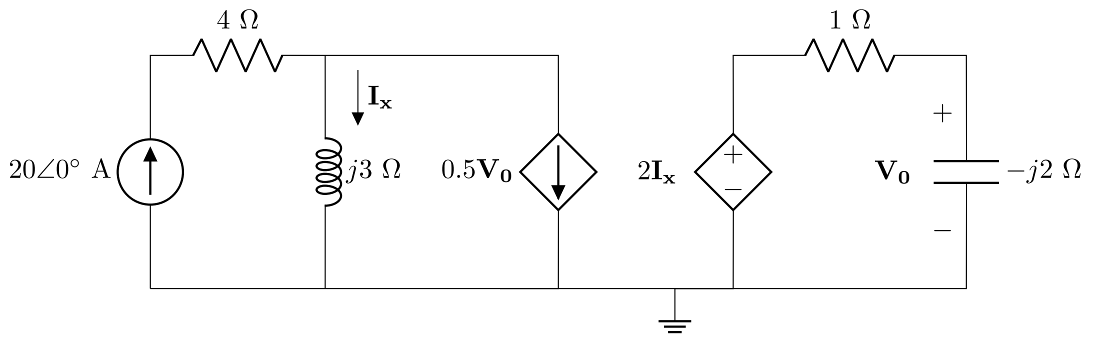
44

$$Z_T = 2 - j2 + \frac{4(4+j6)}{8+j6} = 4.72 - j1.04\,\Omega.$$

45

$$I_o = 160 \cdot \frac{4}{8+j6} = 51.2 - j38.4\,(A)$$

Question 46 [2]



46	Find $\mathbf{V_0}$. [2]

46

$$I_x = 20 - 0.5V_0$$

$$V_0 = 2I_x \cdot \frac{-j2}{1-j2} \Rightarrow$$

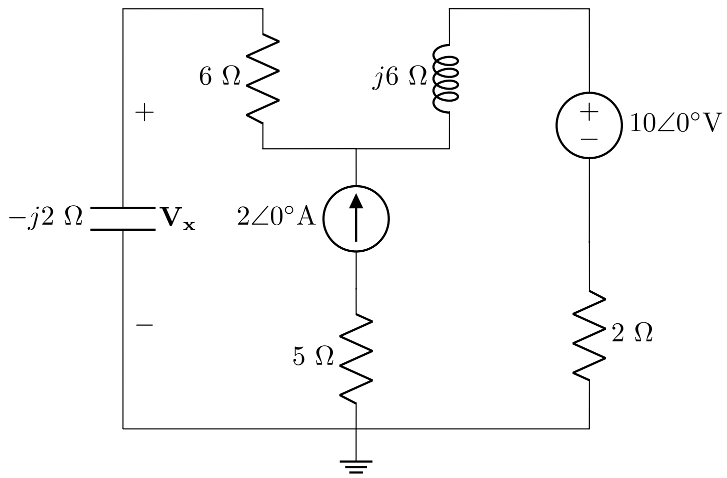
$$(1-j2)V_0 = 2(20-0.5V_0)(-j2)$$

$$= (40 - V_0)(-j2)$$

$$= -j80 + j2V_0$$

$$(1-j4)V_0 = -j80$$

$$\Rightarrow V_0 = \frac{-j80}{1-j4} = 18.82 - j4.706 \text{ V}$$

Questions 47 to 48 [4]	
	
47	Find $\mathbf{V_x^1}$, the exclusive contribution from the voltage source to $\mathbf{V_x}$. [2]
48	Find $\mathbf{V_x^2}$, the exclusive contribution from the current source to $\mathbf{V_x}$. [2]

47: when $2\angle 0^\circ A$ is off, then

$$V_x' = 10 \cdot \frac{-j2}{8+j4} = -1-j2$$

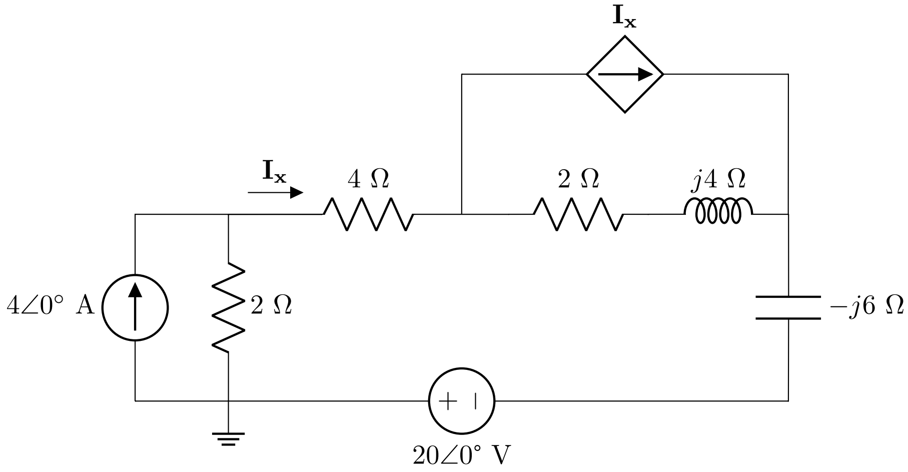
48: when $10\angle 0^\circ$ is off, then

$$V_x^2 = 2 \cdot \frac{2+j6}{8+j4} \cdot (-j2) = 2-j2$$

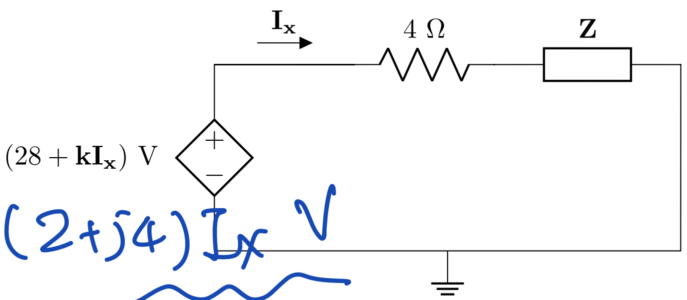
Questions 49 to 50 [4]

Use source transformation to transfer Circuit A to Circuit B.

Circuit A: δ

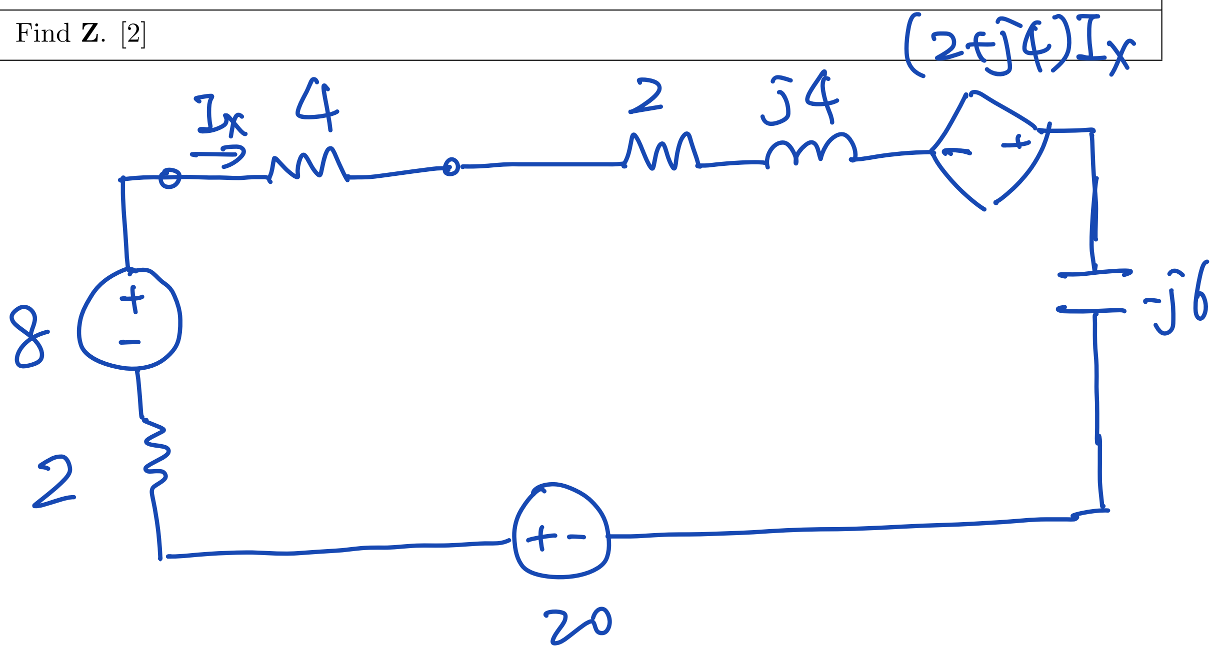


Circuit B:



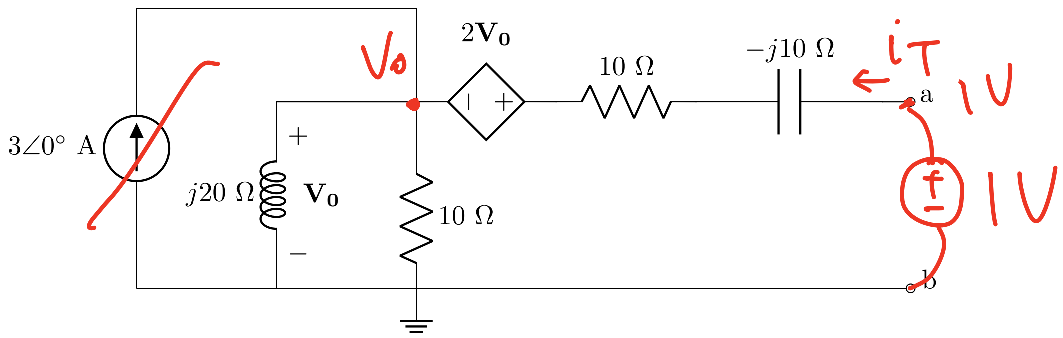
$28 + (2+j4)I_x$ V

49	Find **k**. [2]
50	Find **Z**. [2]



49 $Z = 2 + j4 - j6 + 2 = 4 - j2\ \Omega$.

50 $k = 2 + j4$.

Questions 51 to 52 [4]	
Find the Thévenin equivalent voltage $\mathbf{V_{Th}}$, and Thévenin equivalent impedance $\mathbf{Z_{Th}}$ with respect to terminals $a - b$.	
	
51	$\mathbf{V_{Th}}$ [2]
52	$\mathbf{Z_{Th}}$ [2]

$$V_{Th} = 2V_o + V_o = 3V_o$$

$$V_o = 3 \cdot \frac{j20 \times 10}{10 + j20} = 24 + j12$$

$$\Rightarrow V_{Th} = 72 + j36\ V.$$

$$\frac{V_o}{j20} + \frac{V_o}{10} + \frac{3V_o - 1}{10 - j10} = 0,$$

for Z_{Th}, add $1\angle 0°$ V test voltage. switch off $3\angle 0°$ A.

$$-j0.05\, V_o + 0.1\, V_o + (0.15 + j0.15) V_o = 0.05 + j0.05$$

$$\Rightarrow (0.25 + j0.1) V_o = 0.05 + j0.05$$

$$V_o = \frac{0.05 + j0.05}{0.25 + j0.1} = \frac{5 + j5}{25 + j10} = \frac{1 + j}{5 + j2}$$

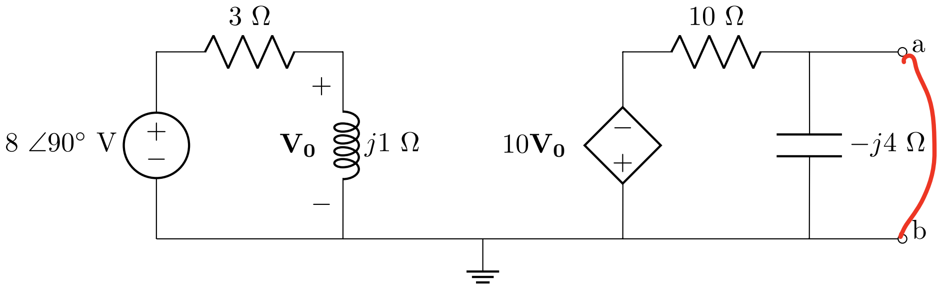
$$i_T = \frac{1-3V_o}{10-j10} = \frac{2-j}{(10-j10)(5+j2)}$$

$$Z_T = \frac{1}{i_T} = \frac{(10-j10)(5+j2)}{2-j}$$

$$= 34+j2\ \Omega.$$

Question 53 [2]

Determine the Norton equivalent current $\mathbf{I_N}$ with respect to terminals $a - b$.



53	$\mathbf{I_N}$ [2]

$$I_N = \frac{-10V_0}{10} = -V_0.$$

$$V_0 = j8 \cdot \frac{j}{3+j} = \frac{-8}{3+j}$$

$$\Rightarrow I_N = \frac{8}{3+j} = 2.4 - j0.8 \text{ A}$$

Total Section 3: [31]

Grand Total : [93]

PRACTICE ANSWER SHEET

Assessment ID:	2	0	2	2	E	B	N	1	1	1	E	0	1

Enter complex numbers on OCR forms:										
5	∠	-	9	0	°				V	
6	.	9	∠	1	5	.	9	°	μ	A
1	2	-	j	8						
j	1	5							Ω	
-	1	.	5	+	j	8	.	2	m	V

Complex numbers can be entered in either **rectangular** or **polar** form.	
Rectangular:	The j is at the start of the imaginary number.
	The j is in the CENTRE of the cell.
	The j does NOT cross or touch cell boundaries!
Polar:	The degree sign ° is directly after the angle value.

Round irrational numbers to a minimum of 4 significant figures.

	Answer	Unit
	SECTION 1	
01	Annual cost (in Rand)=	No unit required
02	$W_T =$	
03	$P_1 =$	
04	$P_2 =$	
05	$i_z =$	
06	$v_z =$	
07	$R_{ab} =$	
08	$R_{cd} =$	
09	$a =$	No unit required
10	$b =$	No unit required
11	$c =$	No unit required
12	$d =$	No unit required
13	$e =$	No unit required
14	$f =$	No unit required
15	$g =$	No unit required
16	$h =$	No unit required
17	$v_x =$	
18	$R_1 =$	
19	$R_2 =$	
20	$I_S =$	
21	$V_{CE} =$	

Round irrational numbers to a minimum of 4 significant figures.

	Answer	Unit
SECTION 2		
22	$v_x^1 =$	
23	$v_x^2 =$	
24	$y =$	
25	$V_x =$	
26	$V_{Th} =$	
27	$R_{Th} =$	
28	$I_N =$	
29	$R_N =$	
30	$V_{TH} =$	
31	$v_1 =$	
32	$i_o =$	
33	$v_o =$	
34	$i_s =$	
35	$v_1 =$	
36	$R_a =$	
37	$v_y =$	
SECTION 3		
38	$C_{eq} =$	
39	$w_T =$	
40	$v(2\text{ ms}) =$	
41	$i(4\text{ s}) =$	
42	The correct statement is:	No unit required
43	$\mathbf{V_T} =$	
44	$\mathbf{Z_T} =$	
45	$\mathbf{I_0} =$	
46	$\mathbf{V_0} =$	
47	$\mathbf{V_x^1} =$	
48	$\mathbf{V_x^2} =$	
49	$\mathbf{k} =$	No unit required
50	$\mathbf{Z} =$	
51	$\mathbf{V_{Th}} =$	
52	$\mathbf{Z_{Th}} =$	
53	$\mathbf{I_N} =$	